

DHA Suffa University
Department of Computer Science
Final Year Project



Capstone Connect
P-2067

Software Design Specifications

Submitted by
Ayushmann Kukreja (CS201245)
Murtaza Mehdi (CS201211)
Abdul Rehman Khan (CS201250)

Supervisor(s)
Urooj Waheed
Yusra Mansoor

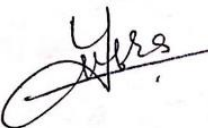
Fall 2023

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Document Sign off Sheet

1.1.1 Document Information

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Name	Role	Signature	Date
Ayushmann Kukreja	Team Lead		27/2/2024
Murtaza Mehdi	Team Member 2		27/2/2024
Abdul rehman Khan	Team Member 3		27/2/2024
Urooj Waheed	Supervisor		27/2/2024
Yusra Mansoor	Co-Supervisor		27/2/2024
Syed Saad Luqman ./ Ali Hassan Sial	Project Coordinator		27/2/2024

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Revision History

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25/2/2024	1.0	Initial version without proper alignment of text for content verification from supervisor	Ayushmann Kukreja Abdul Rehman Khan Murtaza Mehdi
26/2/2024	2.0	Final version with proper alignment	Ayushmann Kukreja Abdul Rehman Khan Murtaza Mehdi

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Definition of Terms, Acronyms, and Abbreviations

Term	Description
SRS	Software Requirement Specification
FYP	Final Year Project
SQL	Structured Query Language
OS	Operating System
IT	Information Technology
WAN	Wide Area Network
LAN	Local Area Network
OAuth	Open Authorization
API	Application Programming Interface
TCP/IP	Transmission Control Protocol/Internet Protocol
FAQs	Frequently Asked Questions

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3 Introduction

3.1 Purpose of Document

This document outlines the design of the Capstone Connect project, a Final Year Project (FYP) portal simulator. It provides detailed information on the expected input, output, classes, and functions, as well as the interactions between these elements to meet the desired requirements. At the conclusion of the document, you will find detailed figures illustrating the relationships and flow of information between different classes.

3.2 Intended Audience

The intended audiences of the SDS document are:

- **The developers** who will design, implement, test, and deploy the CapstoneConnect portal based on the requirements specified in the SDS document.
- **The stakeholders** who will review, approve, and provide feedback on the SDS document and the CapstoneConnect portal.
- **The users** who will review, approve, and provide feedback on the SDS document and the CapstoneConnect portal.

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3.3 Document Convention

Font Family – Times New Roman

Font Sizes:

Top Heading – 20

Middle Headings – 16

Internal Headings – 14

Text - 12

3.4 Project Overview

The Capstone Connect portal aims to address the existing challenges in the academic capstone project lifecycle by creating a centralized and efficient platform. This platform will streamline project selection, enhance collaboration, and improve communication among students, faculty, and industry stakeholders.

3.5 Scope

Project scope definition is the process whereby a project is defined and prepared for execution. It helps to decide on whether or not to proceed with the project. The scope of the CapstoneConnect project is carefully defined to encompass the specific functionalities, features, and objectives intended for development and implementation. The projects in scope elements are outlined below:

1. User Authentication:

Develop secure and user-friendly registration for students, supervisors, and administrators with robust authentication mechanisms.

2. Progress Tracking and Milestone Management:

Design a centralized dashboard for tracking project progress; implement milestone management for FYP lifecycle monitoring.

3. Communication Hub:

Establish a seamless communication hub with messaging and notifications between students and supervisors.

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4. Documentation and Resource Management:

Create a structured system for project document submission and storage; develop tools for resource sharing.

5. Collaborative Tools:

Integrate discussion forums and chat features to encourage knowledge sharing among users.

6. Evaluation and Grading System:

Implement a comprehensive system for supervisors to assess and provide transparent feedback on project submissions.

7. Analytics and Reporting:

Develop analytics for project trends, user engagement, and system performance; provide admin tools for monitoring.

8. Security Measures:

Incorporate robust security measures, conduct regular audits, and ensure compliance with data protection regulations.

9. Scalability and Integration:

Design for scalability, accommodating user and project growth; ensure seamless integration with institutional systems.

10. User Training and Support:

Develop training materials for users and provide ongoing support, including helpdesk services.

11. Continuous Improvement and Feedback Loop:

Establish a structured feedback loop for iterative improvements based on user input and evolving needs.

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4 Design Considerations

4.1 Assumptions and Dependencies

Assumptions:

1. CapstoneConnect portal will use a web-based platform compatible with the chosen software architecture.
2. Compatibility with major browsers like Microsoft Edge, Google Chrome, Mozilla Firefox is ensured.
3. The portal will feature a consistent and intuitive user interface adhering to UX design best practices.
4. A secure and efficient database system will store and retrieve data accurately.
5. The authentication system will be robust and flexible, accommodating various user types and roles.
6. Comprehensive documentation will support development, testing, and maintenance.

Dependencies:

1. Reliance on web server and network availability and performance.
2. Quality and completeness of requirements provided by stakeholders are crucial.
3. User and supervisor cooperation and feedback are essential throughout the project.
4. Integration and compatibility with external systems (institutional systems, email service, discussion forum) are dependencies.
5. Adherence to data protection and security regulations is vital.

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4.2 Risks and Volatile Areas

1. User Feedback

User Feedback is crucial for system improvement, yet it introduces the risk of conflicting requirements. To address this, the system will employ a feedback loop for continuous user monitoring, emphasizing prioritization and validation before integration into the design.

2. Technology Updates

Technology Updates are inevitable and may impact design aspects such as compatibility and security. Mitigation involves using well-established, supported technologies, adhering to coding best practices, and having a contingency plan for potential technology migrations.

3. Security Threats

Security Threats pose a constant risk, necessitating robust measures like encryption, access controls, and regular audits. The system will also have a contingency plan for swift recovery from security breaches.

4. Scalability Challenges

Scalability Challenges may arise with system growth. Mitigation involves designing a scalable architecture, utilizing cloud-based services for flexibility, and closely monitoring and optimizing system performance and resource utilization.

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5 System Architecture

5.1 System Level Architecture

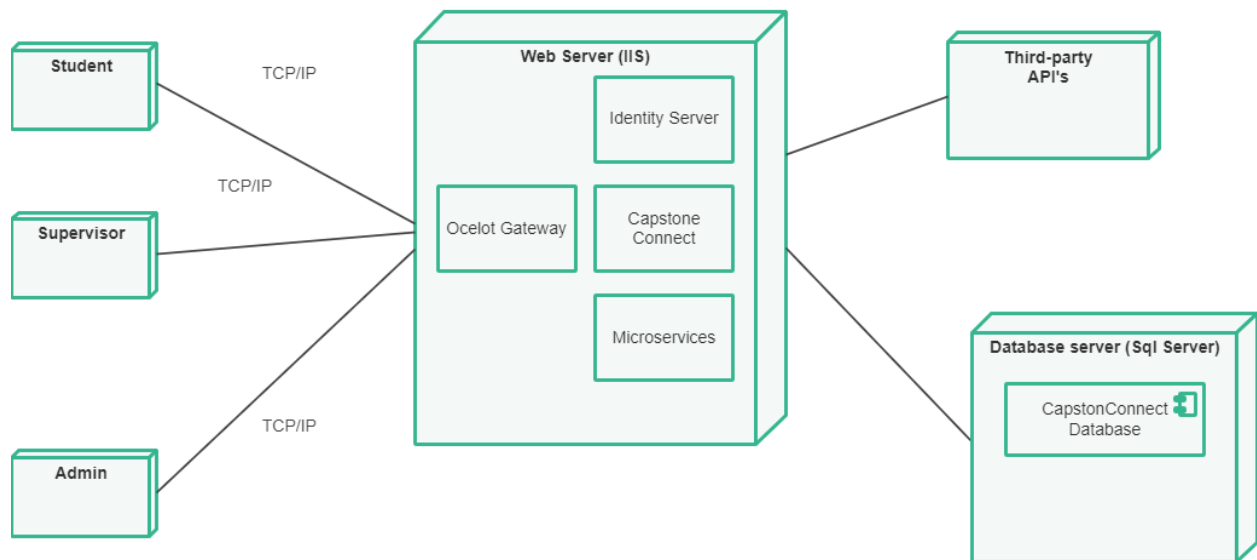


Figure 1: System Level Architecture

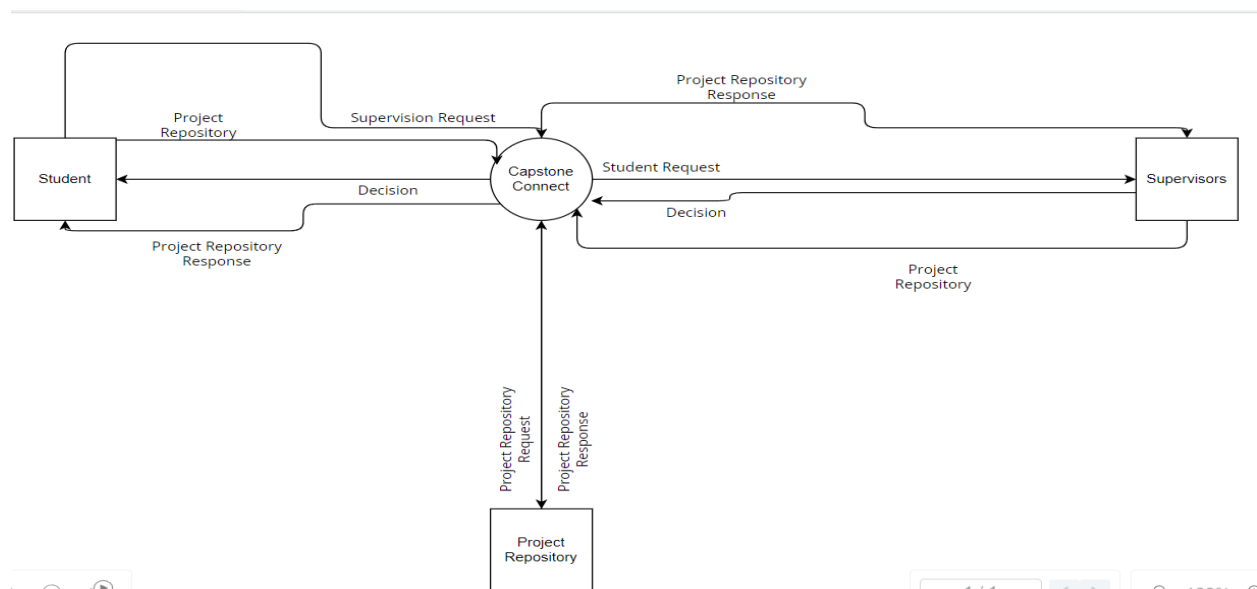


Figure 2: System Level Architecture

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5.2 Software Architecture

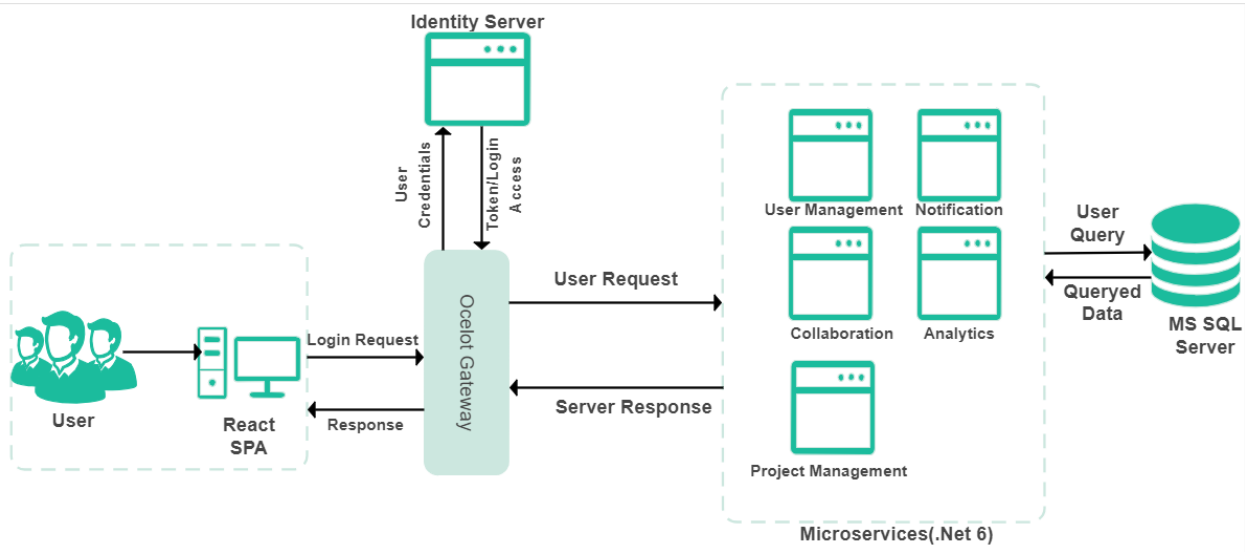


Figure 3: Software Architecture

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6 Design Strategy

1 User-Centered Design:

The portal should be designed with the user in mind, taking into consideration their needs and concerns. This includes making the portal easy to navigate and understand and ensuring that the information is presented clearly and concisely.

2 Security and Privacy:

The portal should be designed with security and privacy in mind, to protect users' personal information and prevent unauthorized access to the portal.

3 Mobile-Friendly:

The portal should be designed to be mobile-friendly, as many users will likely access it using their smartphones or other mobile devices.

4 Feedback and Improvements:

The portal should be designed with the ability to collect feedback and make improvements, based on user feedback and usage data.

5 Ease of Use:

The portal should be designed to be simple and easy to use, with clear Instructions and providing useful information and resources throughout the process.

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7 Detailed System Design

7.1 Design Class Diagram

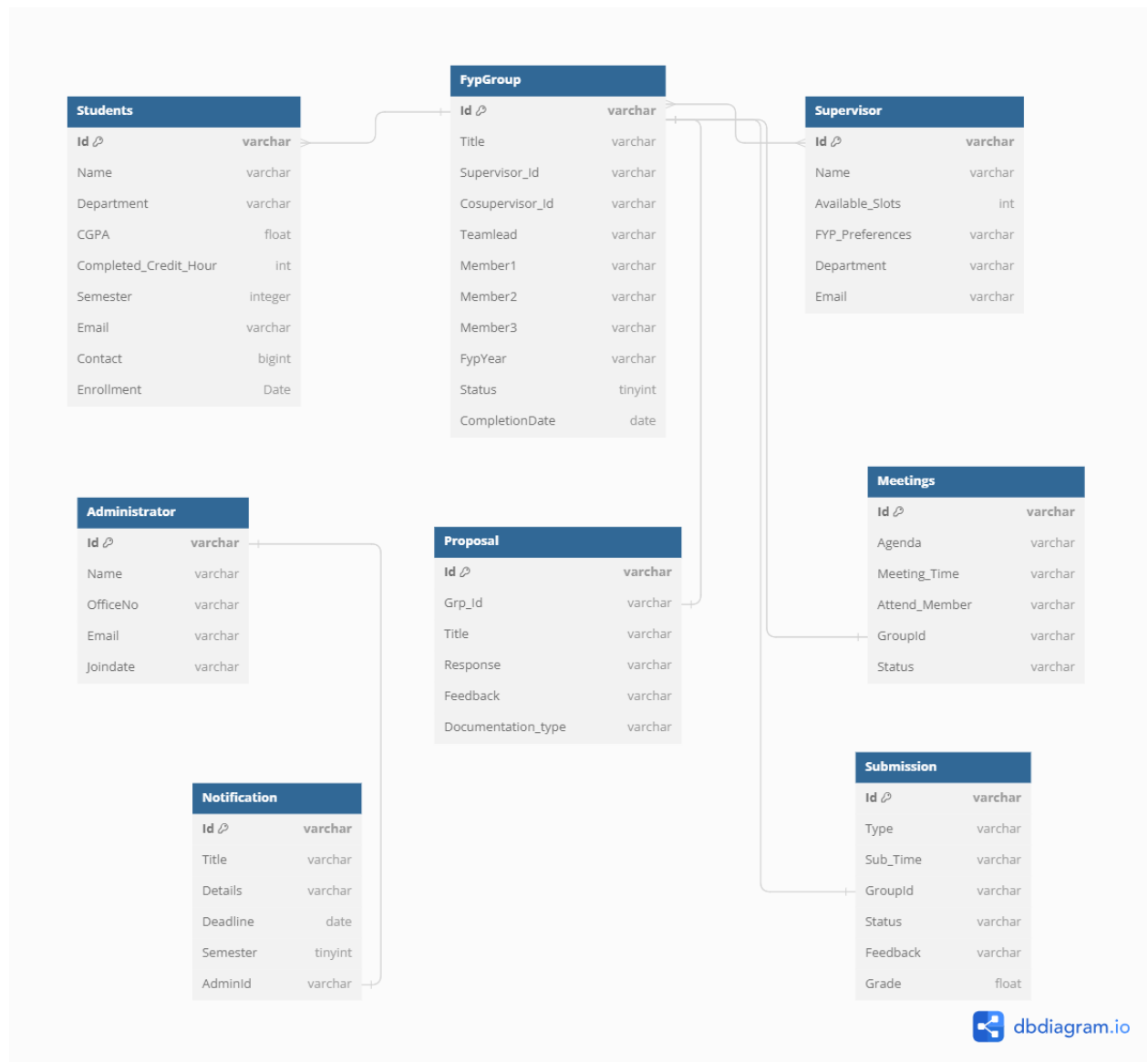


Figure 4: Design Class Diagram

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7.2 Database Design

7.2.1 ER Diagram

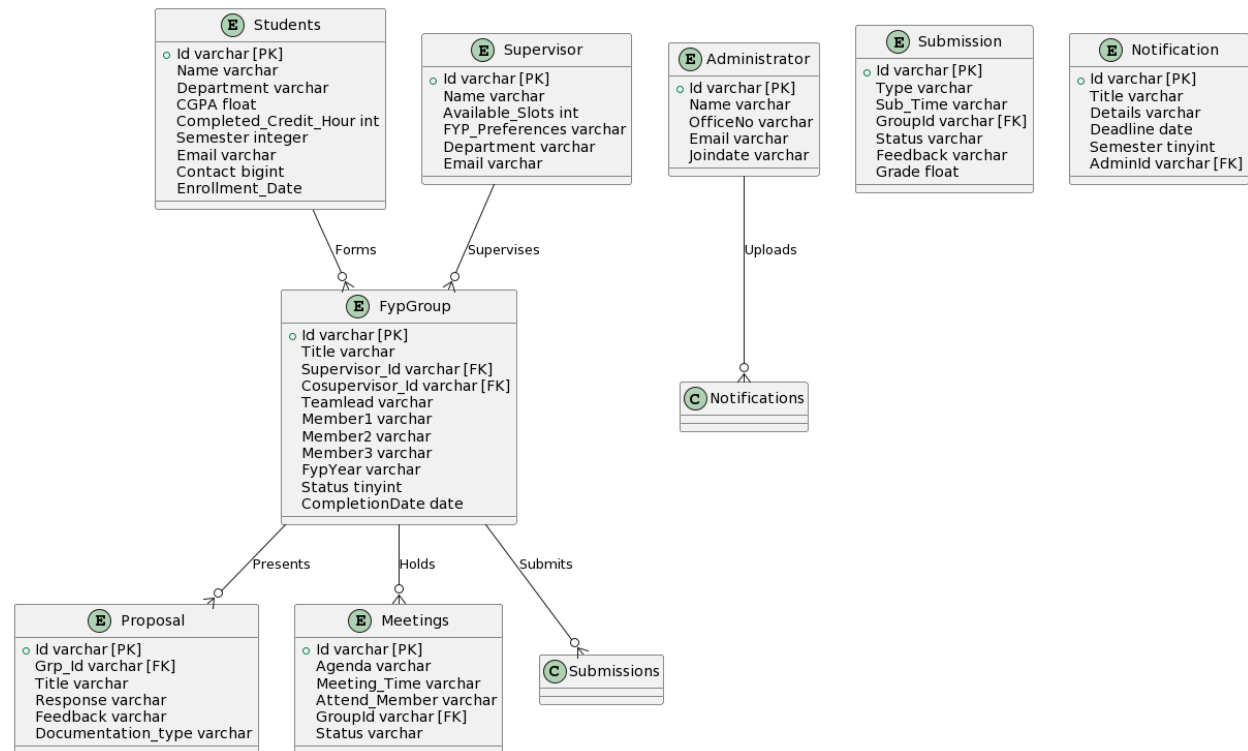


Figure 5: ER Diagram

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7.2.2 Data Dictionary

7.2.2.1 Data 1

Data 1						
Name	Students					
Alias	std					
Where-used/how-used	Primary entity to store student data					
Content description	Notation for representing content.					
Column Name	Description	Type	Length	Null able	Default Value	Key Type
<i>Name</i>	<i>student name</i>	<i>Varchar</i>	<i>100 bytes</i>	<i>No</i>	<i>“ ”</i>	
<i>ID</i>	<i>student id</i>	<i>UUID</i>	<i>50 bytes</i>	<i>No</i>	<i>“ ”</i>	<i>PK</i>
<i>Password</i>	<i>account password</i>	<i>Varchar</i>	<i>30 bytes</i>	<i>Yes</i>	<i>“ ”</i>	
<i>Credit hours</i>	<i>Completed credit hours</i>	<i>int</i>	<i>10 bytes</i>	<i>No</i>	<i>“ ”</i>	
<i>CGPA</i>	<i>Student grade</i>	<i>float</i>	<i>10 bytes</i>	<i>No</i>	<i>2.00</i>	
<i>Semester</i>	<i>current semester</i>	<i>int</i>	<i>10 bytes</i>	<i>No</i>	<i>7</i>	
<i>Email</i>	<i>student email</i>	<i>Varchar</i>	<i>50 bytes</i>	<i>No</i>	<i>“ ”</i>	
<i>Enrollment</i>	<i>Enrollment date</i>	<i>Datetime</i>	<i>50 bytes</i>	<i>No</i>	<i>“ ”</i>	

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7.2.2.2 Data 2

Data 2						
Name		Supervisors				
Alias		spv				
Where-used/how-used		Primary entity to store supervisor data				
Content description		Notation for representing content.				
Column Name	Description	Type	Length	Nullable	Default Value	Key Type
<i>Name</i>	<i>supervisor name</i>	<i>Varchar</i>	<i>100 bytes</i>	<i>No</i>	<i>""</i>	
<i>ID</i>	<i>supervisor id</i>	<i>UUID</i>	<i>50 bytes</i>	<i>No</i>	<i>""</i>	<i>PK</i>
<i>Password</i>	<i>account password</i>	<i>Varchar</i>	<i>30 bytes</i>	<i>Yes</i>	<i>""</i>	
<i>Preferences</i>	<i>area of interest</i>	<i>Varchar</i>	<i>250</i>	<i>No</i>	<i>""</i>	
<i>Slots</i>	<i>Available slots</i>	<i>Varchar</i>	<i>100</i>	<i>No</i>	<i>""</i>	
<i>Email</i>	<i>email</i>	<i>Varchar</i>	<i>50</i>	<i>No</i>	<i>""</i>	

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7.2.2.3 Data 3

Data 3						
Name	FYP					
Alias	fyp					
Where-used/how-used	Primary entity to store current fyp data					
Content description	Notation for representing content.					
Column Name	Description	Type	Length	Null able	Default Value	Key Type
<i>Title</i>	<i>Project name</i>	<i>Varchar</i>	<i>100 bytes</i>	<i>No</i>	<i>""</i>	
<i>ID</i>	<i>Project id</i>	<i>UUID</i>	<i>50 bytes</i>	<i>No</i>	<i>""</i>	<i>PK</i>
<i>Supervisor Id</i>	<i>supervisor of the project</i>	<i>Varchar</i>	<i>50 bytes</i>	<i>No</i>	<i>""</i>	<i>FK</i>
<i>Co Supervisor Id</i>	<i>Completed credit hours</i>	<i>Varchar</i>	<i>50 bytes</i>	<i>Yes</i>	<i>""</i>	<i>FK</i>
<i>Student 1</i>	<i>Group member</i>	<i>Varchar</i>	<i>50 bytes</i>	<i>No</i>	<i>""</i>	<i>FK</i>
<i>Student 2</i>	<i>Group member</i>	<i>Varchar</i>	<i>50 bytes</i>	<i>No</i>	<i>""</i>	<i>FK</i>
<i>Student 3</i>	<i>Group member</i>	<i>Varchar</i>	<i>50 bytes</i>	<i>yes</i>	<i>""</i>	<i>FK</i>
<i>Student 4</i>	<i>Group member</i>	<i>Varchar</i>	<i>50 bytes</i>	<i>Yes</i>	<i>""</i>	<i>FK</i>
<i>Year</i>	<i>Year of completion</i>	<i>int</i>	<i>20 bytes</i>	<i>Yes</i>	<i>""</i>	
<i>Status</i>	<i>Is the project completed?</i>	<i>Tinyint</i>	<i>1 byte</i>	<i>No</i>	<i>""</i>	

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7.2.2.4 Data 4

Data 4						
Name	Submissions					
Alias	subs					
Where-used/how-used	Primary entity to store documents					
Content description	Notation for representing content.					
Column Name	Description	Type	Length	Null able	Default Value	Key Type
<i>ID</i>	<i>document id</i>	<i>UUID</i>	<i>50 bytes</i>	<i>No</i>	<i>""</i>	<i>PK</i>
<i>Project Id</i>	<i>project id</i>	<i>UUID</i>	<i>50 bytes</i>	<i>No</i>	<i>""</i>	<i>FK</i>
<i>Type</i>	<i>Type (image,file etc)</i>	<i>Varchar</i>	<i>200 bytes</i>	<i>No</i>	<i>""</i>	
<i>Feedback</i>	<i>Feedback on the submission</i>	<i>Varchar</i>	<i>200 bytes</i>	<i>Yes</i>	<i>""</i>	
<i>Grade</i>	<i>grade</i>	<i>Float</i>	<i>4 bytes</i>	<i>Yes</i>	<i>""</i>	

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7.2.2.5 Data 5

Data 5						
Name	Admin					
Alias	admin					
Where-used/how-used	Primary entity to store documents					
Content description	Notation for representing content.					
Column Name	Description	Type	Length	Nullable	Default Value	Key Type
<i>ID</i>	<i>notification id</i>	<i>UUID</i>	<i>50 bytes</i>	<i>No</i>	<i>""</i>	<i>PK</i>
<i>Name</i>	<i>Admin name</i>	<i>Varchar</i>	<i>50 bytes</i>	<i>No</i>	<i>""</i>	
<i>Email</i>	<i>Admin email</i>	<i>Varchar</i>	<i>200 bytes</i>	<i>No</i>	<i>""</i>	
<i>Designation</i>	<i>designation inside the institution</i>	<i>Varchar</i>	<i>30 bytes</i>	<i>Yes</i>	<i>""</i>	

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7.2.2.6 Data 6

Data 6						
Name	Meetings					
Alias	mtn					
Where-used/how-used	Primary entity to store documents					
Content description	Notation for representing content.					
Column Name	Description	Type	Length	Nullable	Default Value	Key Type
<i>ID</i>	<i>meeting id</i>	<i>UUID</i>	<i>50 bytes</i>	<i>No</i>	<i>""</i>	<i>PK</i>
<i>Agenda</i>	<i>purpose of the meeting</i>	<i>Varchar</i>	<i>50 bytes</i>	<i>No</i>	<i>""</i>	
<i>Group Id</i>	<i>group organizing the meeting</i>	<i>UUID</i>	<i>50 bytes</i>	<i>No</i>	<i>""</i>	
<i>Status</i>	<i>Ongoing or finished?</i>	<i>tinyint</i>	<i>1 bytes</i>	<i>No</i>	<i>""</i>	

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7.2.2.7 Data 7

Data 7						
Name	Proposal					
Alias	prop					
Where-used/how-used	Primary entity to store documents					
Content description	Notation for representing content.					
Column Name	Description	Type	Length	Nullable	Default Value	Key Type
<i>ID</i>	<i>proposal id</i>	<i>UUID</i>	<i>50 bytes</i>	<i>No</i>	<i>""</i>	<i>PK</i>
<i>Group Id</i>	<i>Group that owns the proposal</i>	<i>UUID</i>	<i>50 bytes</i>	<i>No</i>	<i>""</i>	<i>FK</i>
<i>Feedback</i>	<i>feedback on the proposal</i>	<i>Varchar</i>	<i>200 bytes</i>	<i>No</i>	<i>""</i>	<i>FK</i>
<i>Designation</i>	<i>designation inside the institution</i>	<i>Varchar</i>	<i>30 bytes</i>	<i>Yes</i>	<i>""</i>	

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7.3 Application Design

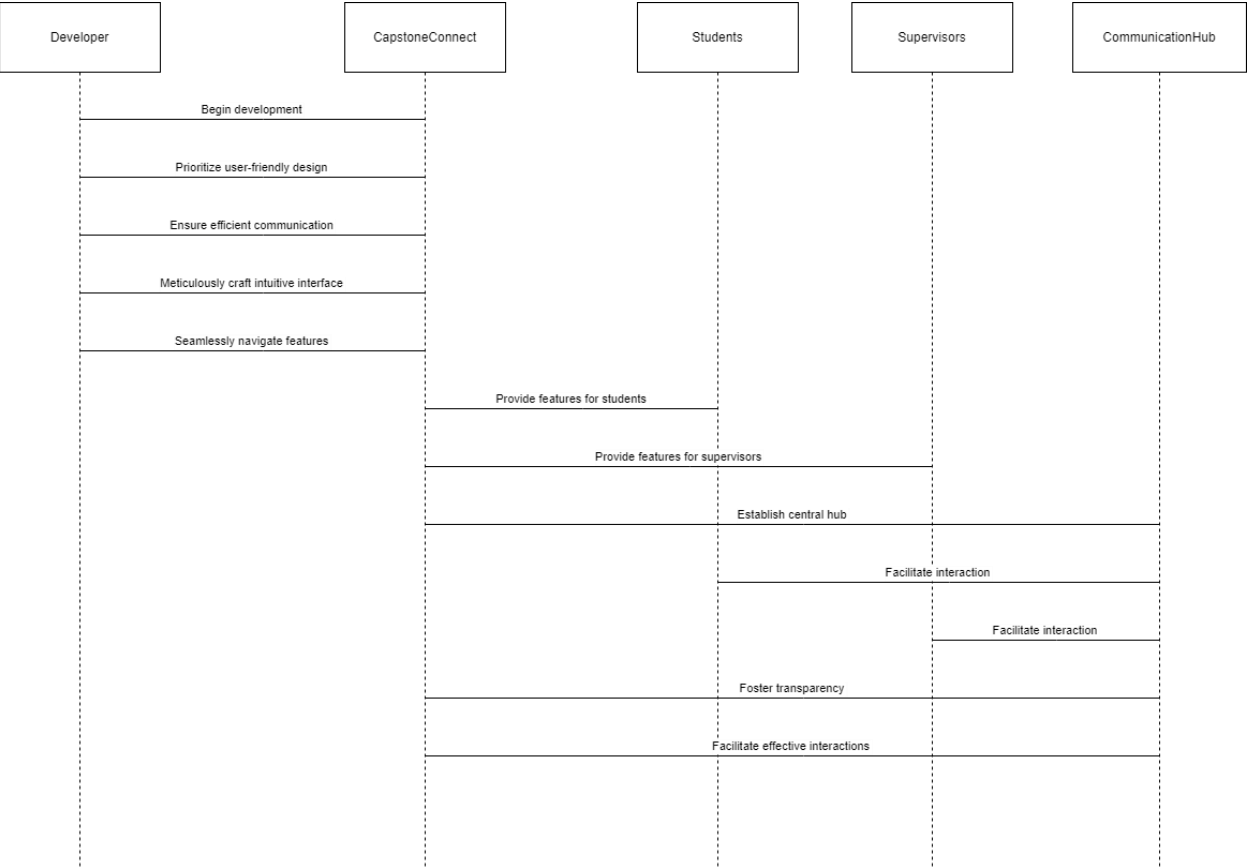


Figure 6: Application Design

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7.3.1 Sequence Diagram

7.3.1.1 <Sequence Diagram - Student>

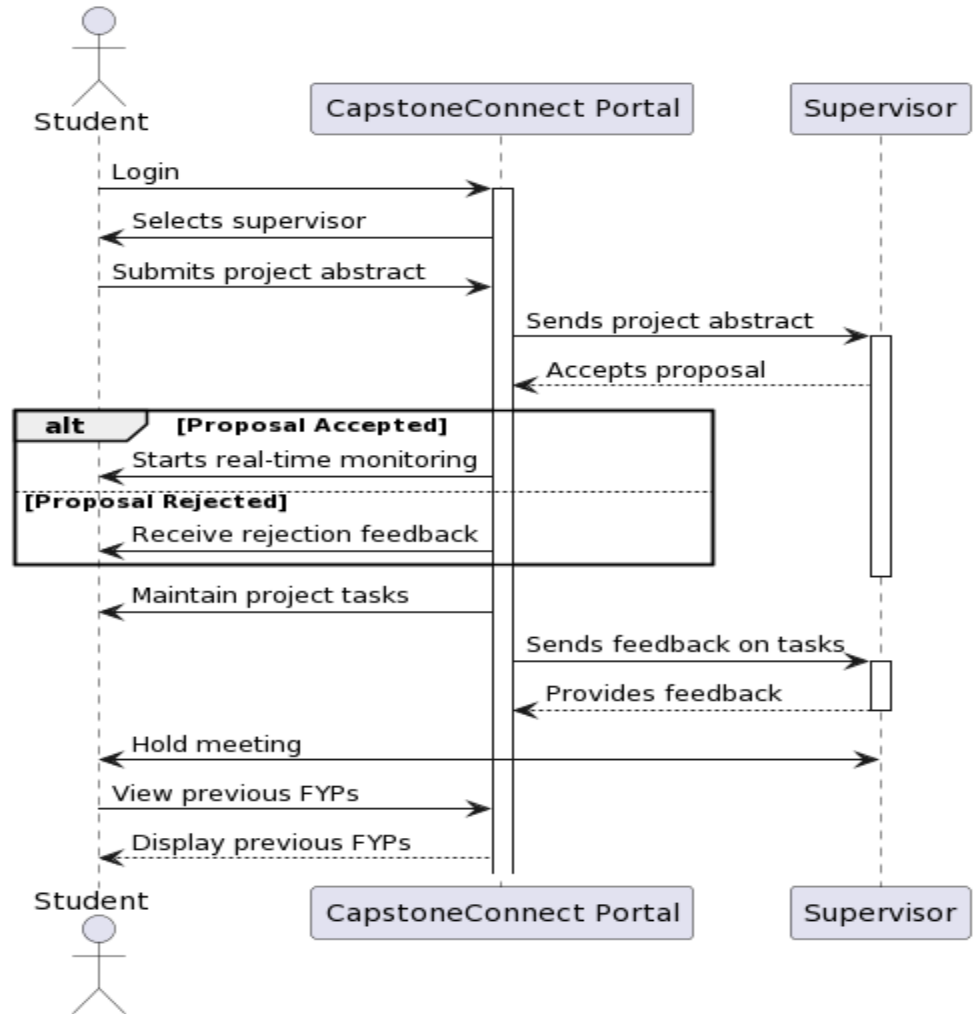


Figure 7: Sequence Diagram: Student

The student interface of CapstoneConnect provides a user-friendly platform for students to seamlessly interact with their supervisors throughout the duration of their final year project (FYP). Upon logging in, students can choose their preferred supervisor, submit project abstracts, and initiate real-time monitoring upon acceptance of their proposal by the supervisor. In case of rejection, supervisors provide constructive feedback to guide students. Throughout the project lifecycle, students can efficiently manage tasks, receive feedback on their progress, schedule meetings with supervisors, and access a repository of previous FYPs for reference, all within the intuitive interface of CapstoneConnect, fostering effective communication and collaboration between students and supervisors.

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7.3.1.2 <Sequence Diagram - Supervisor>

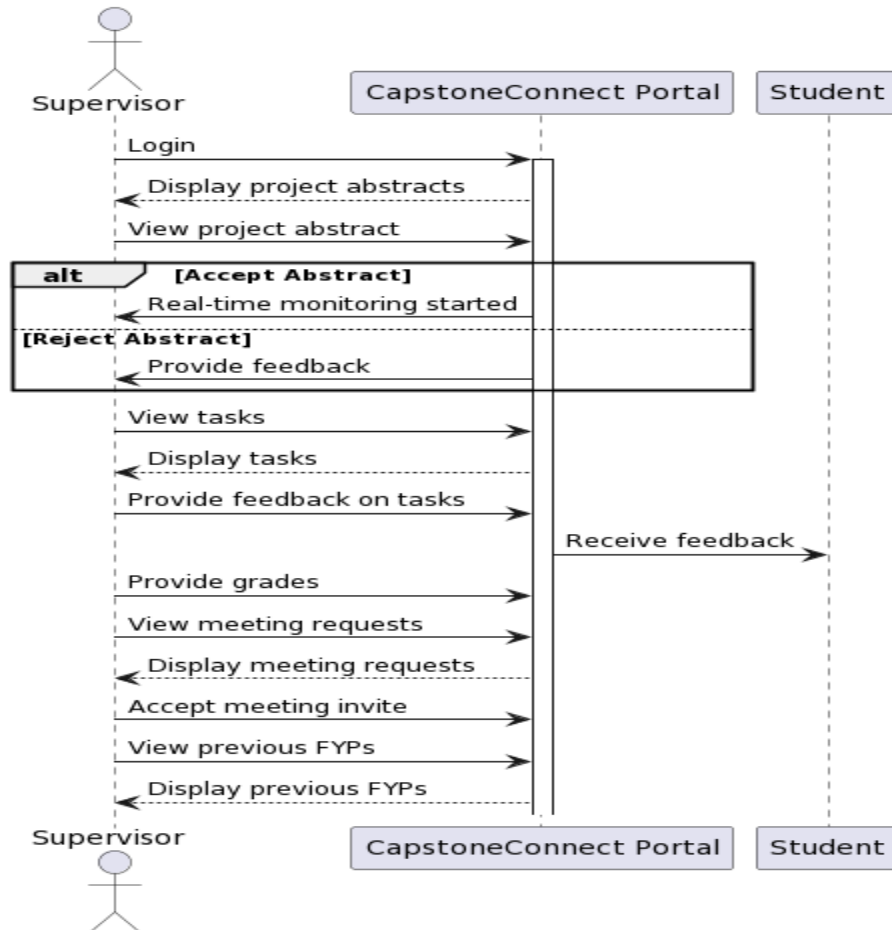


Figure 8: Sequence Diagram: Supervisor

In the CapstoneConnect portal, supervisors begin by logging in, whereupon they can access project abstracts submitted by students. If a supervisor accepts a project abstract, real-time monitoring commences, enabling supervision of the FYP group; otherwise, feedback is provided to guide the students. Supervisors can then view tasks submitted by students, offer feedback, and assign grades on completed tasks. Additionally, they can manage meeting requests, accepting invitations as per their availability. Furthermore, supervisors have the capability to review previous batch FYPs, providing valuable insights and context for their current supervisory responsibilities. This seamless interface facilitates efficient communication and collaboration between supervisors and students throughout the FYP process.

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7.3.1.3 <Sequence Diagram - Administrator>

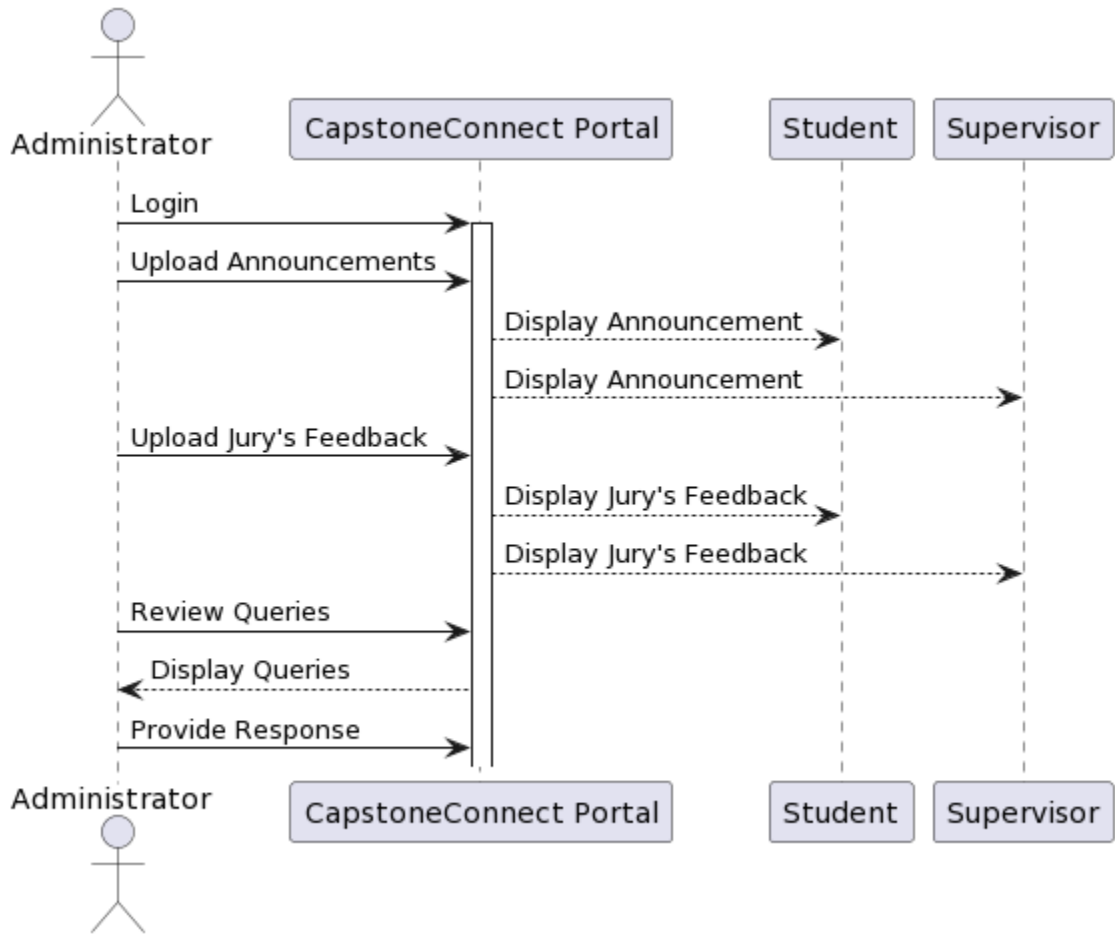


Figure 9: Sequence Diagram Administrator

In the CapstoneConnect portal, the Administrator initiates by logging in, gaining access to crucial functionalities. They upload announcements relevant to the FYP process, ensuring they are promptly displayed to both students and supervisors. Additionally, the Administrator uploads feedback documents from jury members regarding presentations and proposal defenses, which are equally accessible across all interfaces. Furthermore, they review queries submitted by both students and supervisors in real-time, providing timely responses to ensure efficient resolution of concerns and facilitate seamless communication within the CapstoneConnect platform. This interface plays a pivotal role in maintaining transparency, facilitating effective communication, and fostering a conducive environment for the successful completion of FYPs.

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7.3.2 State Diagram

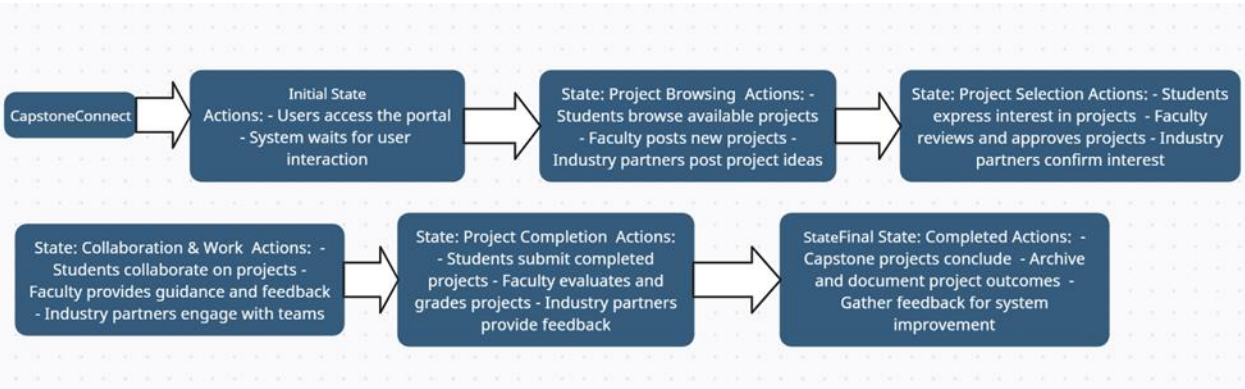


Figure 10: State Diagram

7.4 GUI Design

7.4.1 Login Module

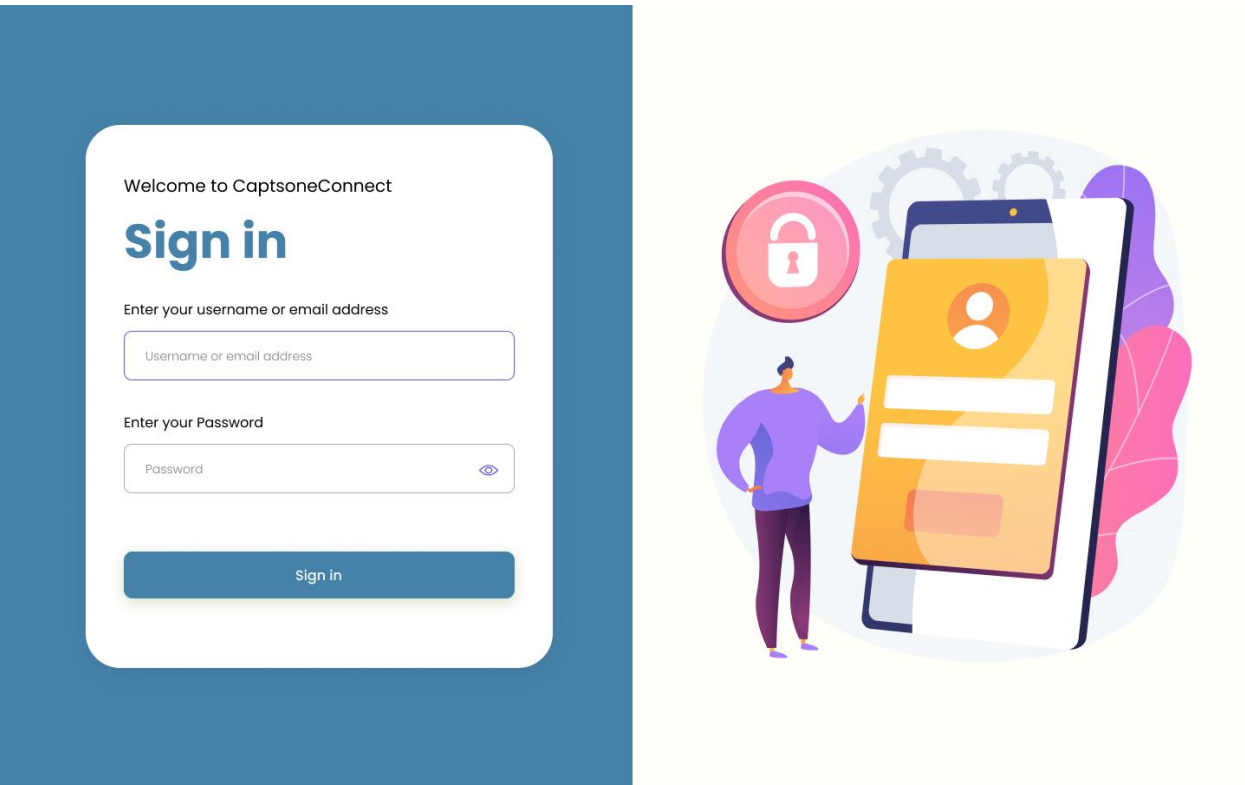


Figure 11: Login

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7.4.2 Available Supervisors List

SUPERVISORS

Total Supervisors : 245 Available : 225 Occupied : 25

Supervisor	Subject	Availability
Supervisor 1	Network Engineering	Available
Supervisor 2	BlockChain	Available
Supervisor 3	Research Paper	Available
Supervisor 4	AI & ML	Available
Supervisor 5	Data Science	Available
Supervisor 6	Robotics	Available

Log Out

Figure 12: Available Supervisors List

7.4.3 Student Dashboard

Capstone Connect

September 4, 2023

Welcome Back, John!
Always stay updated in your student portal

Latest Task

Task	Progress
Task5 - HTML	25%
Task4 - Routing	80%
Task3 - Identifier	20%
Task8 - CSS	80%
Task4 - Scripting	70%
Task9 - RoadMap	30%

Time Spent

Day	Time Spent
Mon	100%
Tue	100%
Wed	100%
Thu	100%
Fri	100%
Sat	100%
Sun	100%

Team Members

Supervisors

Log Out

Figure 13: Student Dashboard

CapstoneConnect	Version: 2.0
Software Design Specifications	Date: 27/2/2024
P-2067	

7.4.4 Projects list for Supervisors

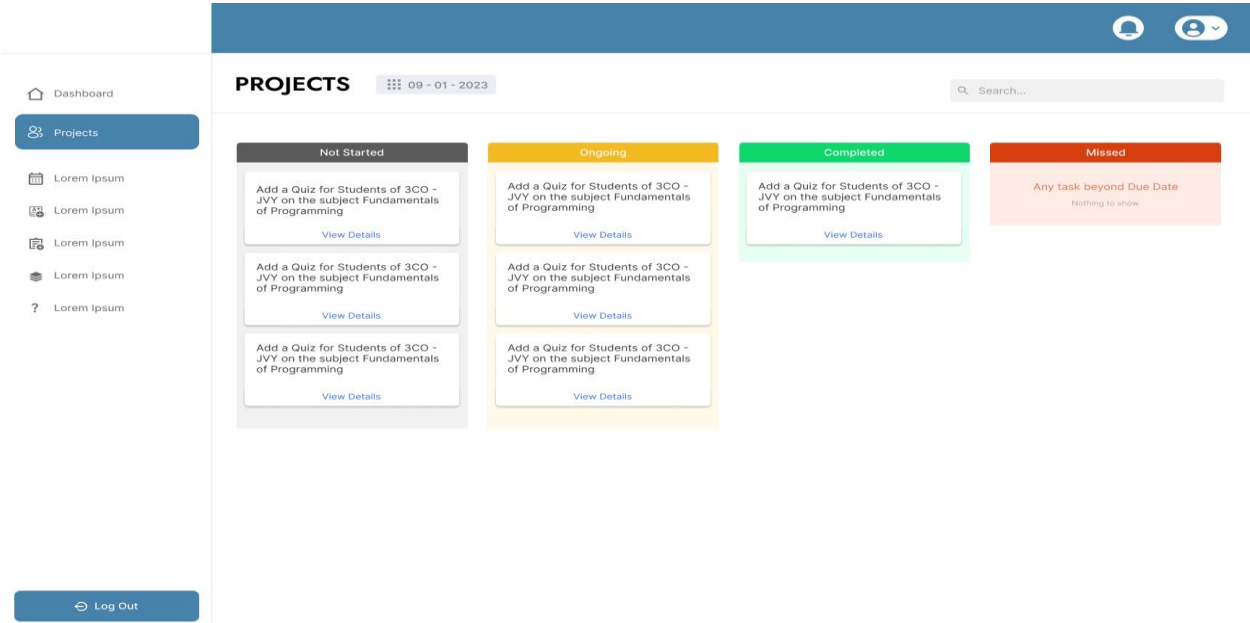


Figure 14: Projects list for Supervisors

7.4.5 FYP Calendar



Figure 15: FYP Calendar

CapstoneConnect	Version: 2.0
Software Design Specifications	Date: 27/2/2024
P-2067	

7.4.6 Previous Project Repository

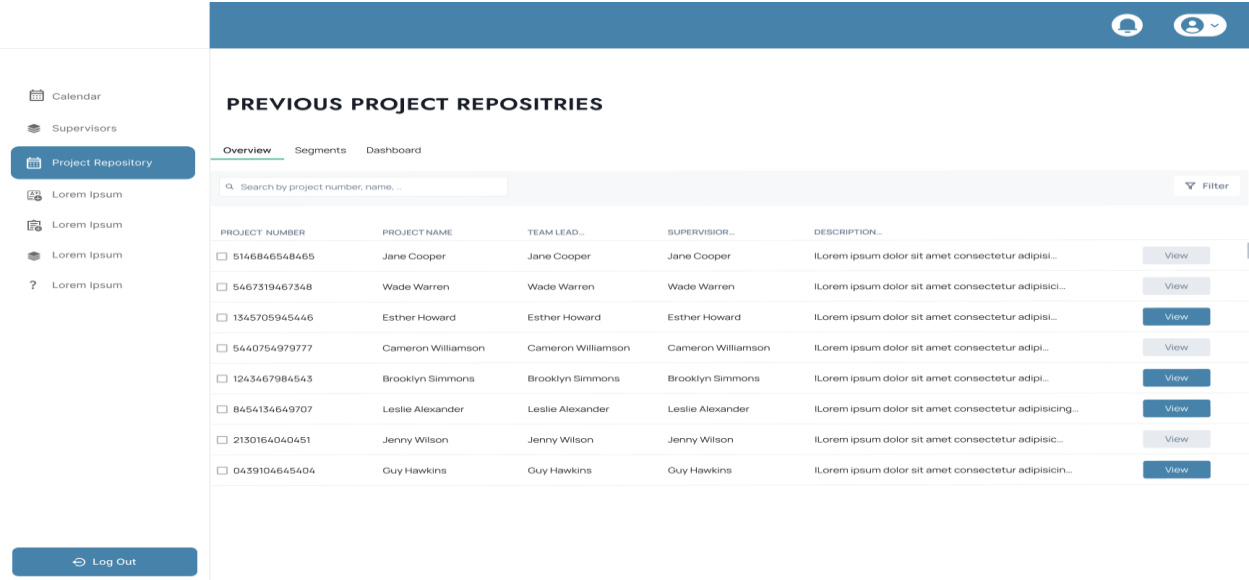


Figure 16: Previous Project Repository

7.4.7 Submission Portal

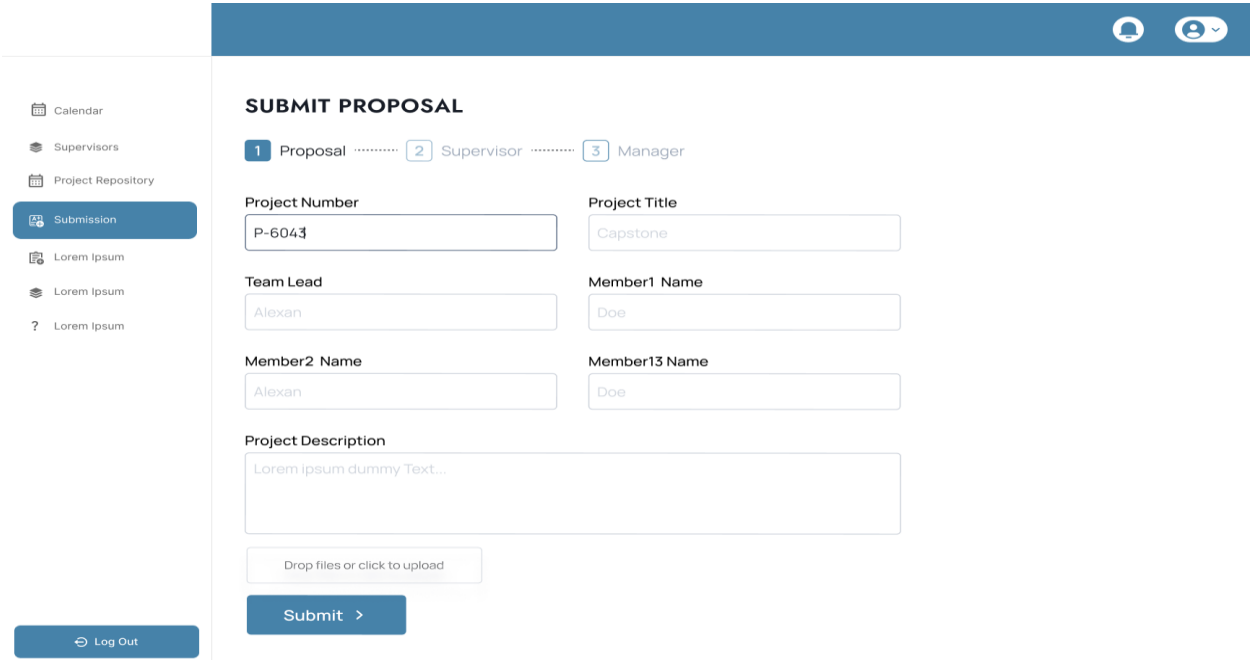


Figure 17: Submission Portal

CapstoneConnect	Version: 2.0
Software Design Specifications	Date: 27/2/2024
P-2067	

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